National University of Computer and Emerging Sciences
Lahore Campus

Probability & Random Processes (EE2011)

Sessional-I Exam

Total Time (Hrs):	1
Total Marks:	60
Total Questions:	6

Date: February 23, 2026

Course Instructor(s)

1. Ms. Aroosa Umair (Course Moderator)
2. Mr. Fasih Gardezi

__________ Roll No __________ Section ________________ Student Signature

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Note: Attempt all questions. Answers must not be given numerically only; explicitly state which probability is being calculated in each part.

CLO #1: Analyze a probabilistic experiment and investigate probabilistic systems

Q1: **[10 marks]**

Two dice are tossed and the magnitude of the difference in the number of dots facing up in the two dice is noted. (a) ***Find*** *the sample space. (b)* ***Find*** *the set A corresponding to the event "magnitude of difference is 3."*

36

6,3
5 2
4 1

Q2: **[10 marks]**

A technology firm is selecting a 3-member project team from a group of 6 employees: 3 Software Engineers (S_1, S_2, S_3) and 3 Hardware Engineers (H_1, H_2, H_3). The selection is done randomly without any restriction. Let event A be that the selected team contains exactly two Software Engineers and one Hardware Engineer. Let event B be that employee $\boxed{S_1}$ is included in the team. **Determine** the probability of event A, also **determine** the probability of event B.

18, 20, 120

$\binom{6}{3}$ $\binom{3}{2}$ L

$\times 3$

$\underline{S_1}$ $\boxed{5}$ $\boxed{4}$

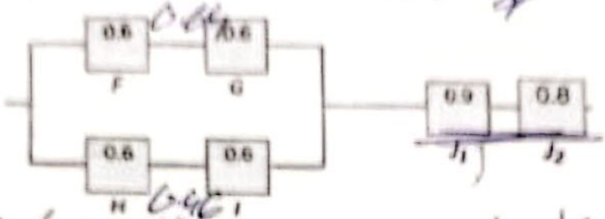
$\cancel{\boxed{5}}$ S_1 _

_ _ $\underline{S_1}$

Q3: 0.43, 0.86. **[10 marks]**

A circuit system is given in Figure 1. Assume the components fail independently.

(a) **Determine** the probability that the entire system works? (b) Given that the system works, **determine** the probability that the component I is not working?



1.28

0.72 ∩ [1 − P(A') P(B')]

P(.72) / 1 − 0.64

Figure 1

444

12/52

Q4: **[10 marks]**

Three cards are drawn in succession without replacement from a standard deck of 52 playing cards. Let event A1 denote that the first card drawn is a face card (Jack, Queen, or King), event A2 denote that the second card drawn is a red number card (2–10 of hearts or diamonds), and event A3 denote that the third card drawn is a black card less than 5. **Find** the probability that the first card is a face card, the second card is a red number card, and the third card is a black card less than 5, given that the cards are drawn without replacement.

$\frac{12}{52} \times \frac{18}{51} \times \frac{8}{50}$

52, 4 types, 13, A, 2→10, K, J, Q

3, 3, 2 2

Q5: **[10 marks]**

A communication system has three channels C1, C2, and C3 used to transmit data packets. Channel C1 carries 40% of the traffic with an error rate of 1%, channel C2 carries 35% of the traffic with an error rate of 2.5% and channel C3 carries 25% of the traffic with an error rate of 4%. If a received packet is found to be erroneous, **determine** the probability that this packet was transmitted through channel C2.

13/3000

Q6: **[10 marks]**

A hospital conducts screening for a viral infection where event A represents that a patient is a smoker and event B represents that a patient tests positive. It is given that 30% patients are smokers and 70% are non-smokers. Among smokers 20% are test positive and among non-smokers, 10% are test positive. **Determine** whether events A and B are mutually exclusive. Also **determine** whether events A and B are independent or dependent.

Good Luck! ☺

1 2 3 4 5 6 7, 8, 9

A, 2, 3, 4, 5, 6, 7, 8, 9, 10